TechPad: After the Parts Arrive

Use this once Hack Club has shipped your kit and PCB. Nothing in it is needed for the submission.

The case was designed with typical part sizes. Real parts vary a little, so check each one below and change only what's different. Every fix is one DXF shape or one number in your FreeCAD file. Then re-export the STEP or STL and print.

1. Measure these with calipers MEASURE

Solder the OLED header and encoder first, so you measure heights as they really sit on the board.

Measure	Value used in the design	If yours is different
OLED: height of the glass top above the PCB top	≈ 5.0 (so it sits flush with the deck)	Lower: no change, it just sits a bit deeper. Higher than 5.1: solder the header lower.
OLED: glass outline, and its distance from header pin 1	31 × 12, starting 4.7 mm right of the pins (X 18.7 to 49.7, Y 70 to 82)	Move or resize the frame's CUT_OLED_WINDOW to the glass + 1 mm at each end.
OLED: module board size	38 × 12 (X 12.7 to 50.7)	Resize the plate's CUT_OLED_MODULE to the board size + 0.5 on each side.
Encoder: height of the square metal body above the PCB	≈ 6.5 (top at Z 8.1, inside the frame's 2 mm pocket)	Taller: make POCKET_ENCODER_UNDERSIDE deeper (body top + 0.5). Bushing wider than 7.5: grow CUT_ENCODER_HOLE.
Encoder: shaft type and height	D-shaft Ø6, 4.5 across the flat, top at Z ≈ 21.6	Knurled shaft: use a plain Ø6.1 bore with a 0.2 mm slit. Different flat: move the D-flat box.
XIAO: PCB back to the centre of the USB-C opening	≈ 2.6 (USB centre Z ≈ −2.6)	Move the USB_SLOT so its centre matches the USB-C centre (it is 6 tall).
Heat-set insert outer diameter and length	5.0 × 4.0 → hole Ø4.7 × 4.0	Hole = insert OD − 0.3.
Screw length and head	M3×16, socket head Ø5.5 × 3.0	Counterbore = head Ø + 0.7; depth = head height.

2. Fit tests (about 30 minutes of printing)

1. **Plate corner:** in FreeCAD, duplicate the plate (Edit → Duplicate selection). Make a **Part → Cube** 30 × 29 × 6 at position −3, −3, 1. Select the copy, then the cube, and run **Part → Boolean → Intersection** (Common). You get one corner with 1 switch hole, 1 boss and 1 screw hole.
2. **Frame corner:** the same with a duplicate of the frame and a Cube 15 × 15 × 7 at position −3, −3, 6.5. You get a corner with 1 insert hole.
3. Export both as STL and print them with the settings in section 4.
4. Check that a switch clicks in firmly and the insert goes in straight. Loose switch: shrink the 14.0 holes by 0.05. Tight: grow them by 0.05. Then reprint the plate corner.

5. Stack the plate corner on the frame corner with a screw through both. They should sit flat together.

3. Knob fit

Print the knob on its own and push it onto the shaft. If it's loose, shrink the bore by 0.05 and reprint. If it won't go on, grow it by 0.05.

4. Print settings Likely

Part	Orientation	Settings
Top frame	**Face-down** (top surface on the bed). A textured PEI sheet gives a matte top.	PLA or PETG, 0.2 mm layers, 3 walls, 20% infill, no supports.
Switch plate	Top face on the bed, bosses pointing up	Same, but 100% infill or 5 walls so the plate stays stiff.
Bottom tray	Floor on the bed	Same. The counterbores face the bed, which is fine. The USB slot top bridges 12 mm, which prints cleanly.
Knob	Top face on the bed	0.12 to 0.16 mm layers for a smooth finish.

Colour: matte charcoal or black, to match the design sheet. Optional: print the plate in a slightly lighter grey, so it shows as a thin stripe on the side of the case.

5. Assembly order

1. Solder everything except the switches: diodes, LEDs from the back, the XIAO on the back, the OLED (see the next step) and the encoder. Flash the firmware and test the LEDs and OLED now, while it's easy to fix things (section 6).
2. **Screen height:** when you solder the OLED module, don't push it all the way down on its pins. Let it sit up to 3.5 mm higher than normal (a 3 mm drill bit makes a handy spacer), so the glass ends up about 3 mm below the frame top. The frame has a pocket for this, and the higher glass is much easier to read past the top row of keys. Check that at least 1.5 mm of each pin still goes through the PCB. Likely
3. Heat-set the 4 inserts into the underside of the top frame, soldering iron at about 220 °C for PLA. Press them flush and straight.
4. Clip all 12 switches into the switch plate.
5. Lower the PCB onto the switch pins. The encoder and the OLED module should go up through their plate cuts, and the 4 bosses should sit just above the PCB. Solder the 24 switch pins.
6. Put the PCB-and-plate into the tray. The plate rests on the tray walls. Put the frame on top.
7. Drive the 4 M3×16 screws from underneath, through the tray, PCB and plate into the inserts, until snug. Don't over-tighten, or the plastic strips.
8. Fit the keycaps and the knob. To open it again later, undo the 4 screws.

6. Flash and test the firmware

1. Before the switches go in: hold the XIAO's **BOOT** button, plug in USB-C, and let go. A drive called **RPI-RP2** appears.
2. Drag **firmware.uf2** onto it. The XIAO restarts by itself.

3. Check the OLED shows text and the LEDs light. Short each switch's two pads with tweezers to test the keys before you solder the switches.
4. Untested firmware may need fixes. If something doesn't work, note what you see and send it to Claude.